

AI-Powered Phishing Detector

High-Level Design and Used Tools

1. High-Level Design

System Overview:

The system analyzes email content and classifies it as phishing or legitimate using a machine learning pipeline.

System Components:

- Input Layer: Accepts email text
- Preprocessing: Cleans and normalizes text
- Feature Extraction: Converts text to numerical form using TF-IDF
- ML Model: Classifies email (phishing or legitimate)
- Output: Displays result

Functional Flow:

Input → Preprocessing → Feature Extraction → Model → Output

2. Used Tools

- Python: Main programming language
- Scikit-learn: ML models and TF-IDF
- Pandas: Data handling
- NumPy: Numerical processing
- NLTK: Text preprocessing
- Flask (optional): Web interface

3. Conclusion

The system uses a simple, modular pipeline with effective tools for accurate phishing detection.